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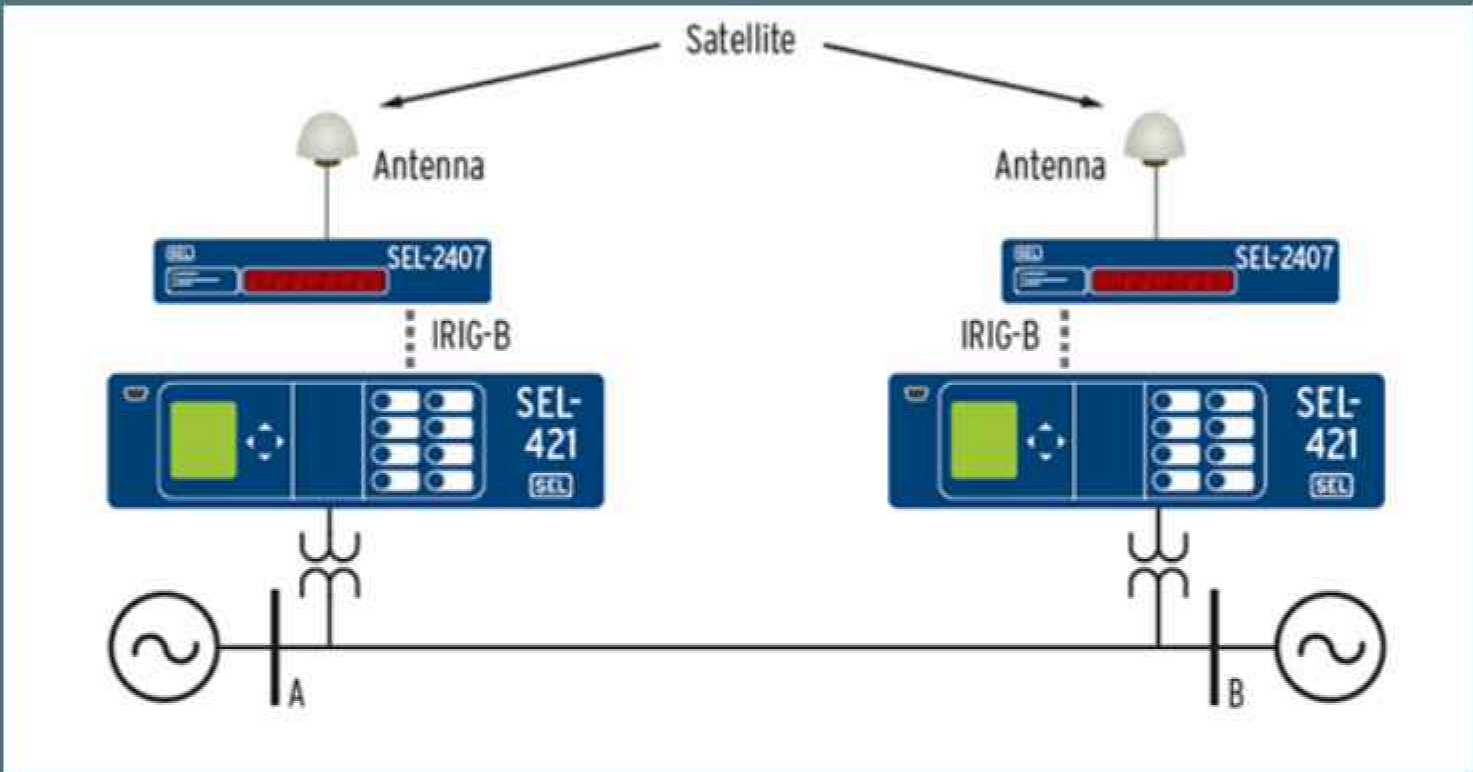
Welcome to TechInnovate, your go-to source for the latest developments, breakthroughs, and cutting-edge technologies in the dynamic field of Electrical and Electronics Engineering. In an era driven by technological evolution, our magazine aims to be the beacon guiding professionals, enthusiasts, and curious minds through the intricate landscape of electrical and electronics innovation.

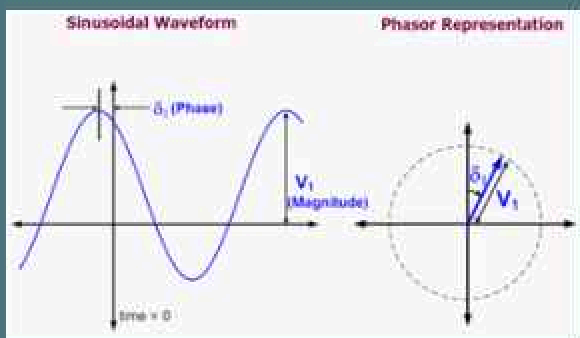
Join us on the journey of innovation and discovery at TechInnovate as we navigate the ever-evolving landscape of Electrical and Electronics Engineering. Together, let's illuminate the path towards a future shaped by technological brilliance and ingenuity.

SYNCHROPHASOR TECHNOLOGY



SYNCHRONIZED PHASOR MEASUREMENTS (SYNCHROPHASORS) TO PROVIDE A REAL-TIME MEASUREMENT OF ELECTRICAL QUANTITIES FROM ACROSS A POWER SYSTEM. THESE MEASUREMENTS CAN BE USED FOR CONTROL, MEASUREMENT, AND ANALYSIS OF THE POWER SYSTEM. APPLICATIONS INCLUDE VALIDATION OF SYSTEM MODELS, MEASUREMENT OF STABILITY MARGINS, MAXIMIZING STABLE SYSTEM LOADING, ISLANDING DETECTION, AND SYSTEM-WIDE DISTURBANCE MONITORING.

WIDESPREAD AVAILABILITY OF A STANDARDIZED TIME REFERENCE PROVIDES THE FOUNDATION FOR SYNCHROPHASOR TECHNOLOGY. BECAUSE THE ELECTRICAL SYSTEM IS CONSTANTLY MOVING, A PHASOR MEASUREMENT MEANS NOTHING WITHOUT COMPARING IT TO ANOTHER PHASOR MEASUREMENT TAKEN AT THE SAME EXACT MOMENT IN TIME. WITH THE INTRODUCTION OF VERY ACCURATE SATELLITE CLOCKS, SYNCHROPHASOR MEASUREMENTS CAN BE TAKEN OVER A LARGE GEOGRAPHIC AREA USING MULTIPLE DEVICES, SUCH AS SEVERAL SEL-421, SEL-451 AND SEL-734 PROTECTION, AUTOMATION, AND CONTROL SYSTEMS.



THAMARIKANNAN V
I YEAR EEE

energy saver

ELECTRICAL ENERGY SAVING

strategies and technologies for electrical energy saving:

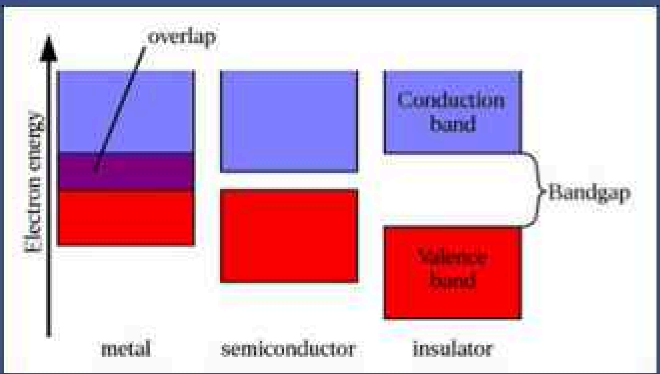
1. Energy-Efficient Lighting:
   - Transition to energy-efficient LED lighting, which consumes significantly less electricity compared to traditional incandescent bulbs.
   - Implement occupancy sensors and smart lighting controls to ensure that lights are only on when needed.
2. Appliance Efficiency:
   - Choose energy-efficient appliances and electronics by looking for the ENERGY STAR label.
   - Unplug electronic devices and chargers when not in use to prevent "phantom" or standby power consumption.
3. HVAC Systems Optimization:
   - Regularly maintain and clean heating, ventilation, and air conditioning (HVAC) systems to ensure optimal efficiency.
   - Install programmable thermostats to control heating and cooling based on occupancy and schedule.
4. Energy-Efficient Motors and Drives:
   - Upgrade motors to high-efficiency models, especially in industrial applications.
   - Implement variable frequency drives (VFDs) to control motor speed and match it to the required load.
5. Power Factor Correction:
   - Address power factor issues to optimize the use of electrical power in industrial and commercial settings, reducing wasted energy.
6. Renewable Energy Integration:
   - Install on-site renewable energy sources such as solar panels or wind turbines to generate clean electricity and offset consumption from the grid.
7. Employee Awareness and Engagement:
   - Educate employees about energy-saving practices and encourage their participation in energy conservation initiatives.
8. Demand Response Programs:
   - Participate in demand response programs that allow organizations to adjust their electricity usage during peak demand periods, often in exchange for incentives.
9. Energy Management Systems (EMS):
   - Deploy EMS to monitor, analyze, and optimize energy consumption in real-time, allowing for proactive decision-making.

WIDE-BANDGAP (WBG) SEMICONDUCTORS

BALASANDHIYA K I YEAR EEE

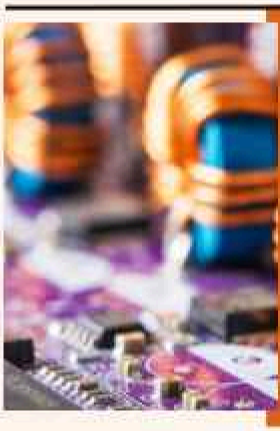
Wide-bandgap semiconductors (also known as WBG semiconductors or WBGSs) are semiconductor materials which have a larger band gap than conventional semiconductors. Conventional semiconductors like silicon have a bandgap in the range of 0.6 – 1.5 electronvolt (eV), whereas wide-bandgap materials have bandgaps in the range above 2 eV. Generally, wide-bandgap semiconductors have electronic properties which fall in between those of conventional semiconductors and insulators.

Wide-bandgap semiconductors permit devices to operate at much higher voltages, frequencies, and temperatures than conventional semiconductor materials like silicon and gallium arsenide. They are the key component used to make short-wavelength (green-UV) LEDs or lasers, and are also used in certain radio frequency applications, notably military radars. Their intrinsic qualities make them suitable for a wide range of other applications, and they are one of the leading contenders for next-generation devices for general semiconductor use.

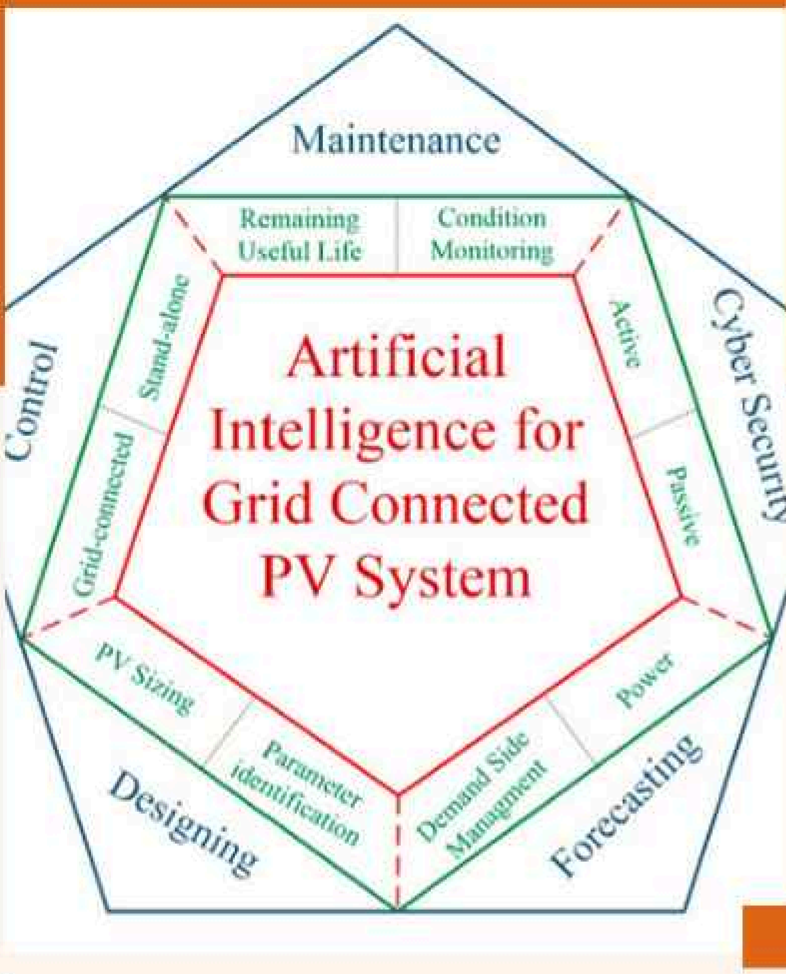


The wider bandgap is particularly important for allowing devices that use them to operate at much higher temperatures, on the order of 300 °C. This makes them highly attractive for military applications, where they have seen a fair amount of use. The high temperature tolerance also means that these devices can be operated at much higher power levels under normal conditions. Additionally, most wide-bandgap materials also have a much higher critical electrical field density, on the order of ten times that of conventional semiconductors. Combined, these properties allow them to operate at much higher voltages and currents, which makes them highly valuable in military, radio, and power conversion applications. The US Department of Energy believes they will be a foundational technology in new electrical grid and alternative energy devices, as well as the robust and efficient power components used in high-power vehicles from plug-in electric vehicles to electric trains. Most wide-bandgap materials also have high free-electron velocities, which allows them to work at higher switching speeds, which adds to their value in radio applications. A single WBG device can be used to make a complete radio system, eliminating the need for separate signal and radio-frequency components, while operating at higher frequencies and power levels.

AI AND MACHINE LEARNING IN POWER ELECTRONICS



AJEETHA A M,
IV YEAR EEE



Artificial intelligence (AI) and machine learning (ML) in power electronics build on the existing foundation of digital power and represent the next step in the evolution of power converter design, control, and optimization. Just as digital power enables more complex control algorithms than analog control techniques, AI and ML will allow even more complex and dynamic non-linear control surfaces to enhance efficiency, reliability predictions, and health monitoring in power converters. This FAQ begins by briefly looking at the growing interest in AI/ML for power electronics applications, it then presents two examples of using AI/ML, one for grid-tied solar installations and one for motor drives, and it closes with a review of some of the challenges to the wide-spread application of AI/ML in power electronics.

Power converter designers and power semiconductor companies are actively developing AI/ML technologies. Research and development activity related to AI in power electronics is exploding (Figure 1). The application of AI/ML in power electronics is different from its use in more established areas such as image classification, speech recognition, etc. Power converter design, optimization of control loops, and preventative maintenance are three key areas where AI/ML are being used.

AI is being used to improve PV systems' design, forecasting, control, and maintenance, resulting in improved return on investment for system owners and operators. In addition, AI is being employed to improve the cyber security of grid-tied PV systems increasingly connected to the cloud F

Grid-tied solar and AI

A convergence of developments, including lower-cost, high-performance computing resources, improved AI tools, and increasing amounts of relevant data sets, are driving the growing use of AI/ML in photovoltaic (PV) systems, especially in grid-tied PV.

ENERGY CONSERVATION

Energy conservation involves the efficient and responsible use of energy resources to reduce consumption and minimize environmental impact. It is a critical aspect of sustainable living and is applicable in various sectors, including residential, commercial, and industrial settings. Here are some key strategies and practices for energy conservation:

- Efficient Technology Adoption
- Smart Habits and Practices
- Renewable Energy Integration

HEMANTH N, IV year EEE

Energy conservation is a collective effort that involves individuals, businesses, and governments working together to reduce energy consumption, lower greenhouse gas emissions, and build a more sustainable future. It requires a commitment to adopting energy-efficient technologies, changing behaviors, and continually seeking ways to minimize energy waste.

Key strategies and practices for energy conservation:

Key strategies and practices for energy conservation:

1. **Energy-Efficient Lighting:**
   - Use energy-efficient lighting solutions, such as LED bulbs, which consume less energy and have a longer lifespan compared to traditional incandescent bulbs.
2. **Appliance Efficiency:**
   - Choose energy-efficient appliances and electronics with high ENERGY STAR ratings to reduce energy consumption. Look for energy labels when purchasing new equipment.
3. **Smart Thermostats:**
   - Install programmable or smart thermostats to optimize heating and cooling systems, adjusting temperatures based on occupancy and time of day to reduce unnecessary energy use.
4. **Insulation and Sealing:**
   - Improve insulation in buildings to minimize heat loss during winter and prevent heat gain during summer. Seal gaps and cracks in doors, windows, and walls to enhance energy efficiency.
5. **Energy-Efficient Windows:**
   - Install energy-efficient windows that provide better insulation, reducing the need for heating or cooling and lowering energy bills.
6. **Renewable Energy Sources:**
   - Consider incorporating renewable energy sources such as solar panels, wind turbines, or geothermal systems to generate clean and sustainable energy on-site.
7. **Water Heater Efficiency:**
   - Set water heaters at optimal temperatures and insulate hot water pipes to reduce heat loss. Consider upgrading to a more energy-efficient water heater if necessary.
8. **Energy-Efficient Landscaping:**
   - Plant trees strategically to provide shade and reduce cooling needs in the summer. Use landscaping to create windbreaks and reduce the impact of cold winds in the winter.
9. **Power Strips and Unplugging Devices:**
   - Use power strips to easily turn off multiple devices at once, preventing standby power consumption. Unplug chargers and electronic devices when not in use.
10. **Energy Audits:**
    - Conduct energy audits for homes and businesses to identify areas of inefficiency and implement targeted improvements. Professional auditors can provide recommendations for energy-saving measures.
11. **Smart Appliances and Home Automation:**
    - Invest in smart appliances and home automation systems that allow for better control and scheduling of energy-consuming devices, optimizing energy use.

UGESH V R, IV year EEE

SOLID-STATE BATTERIES FOR EV

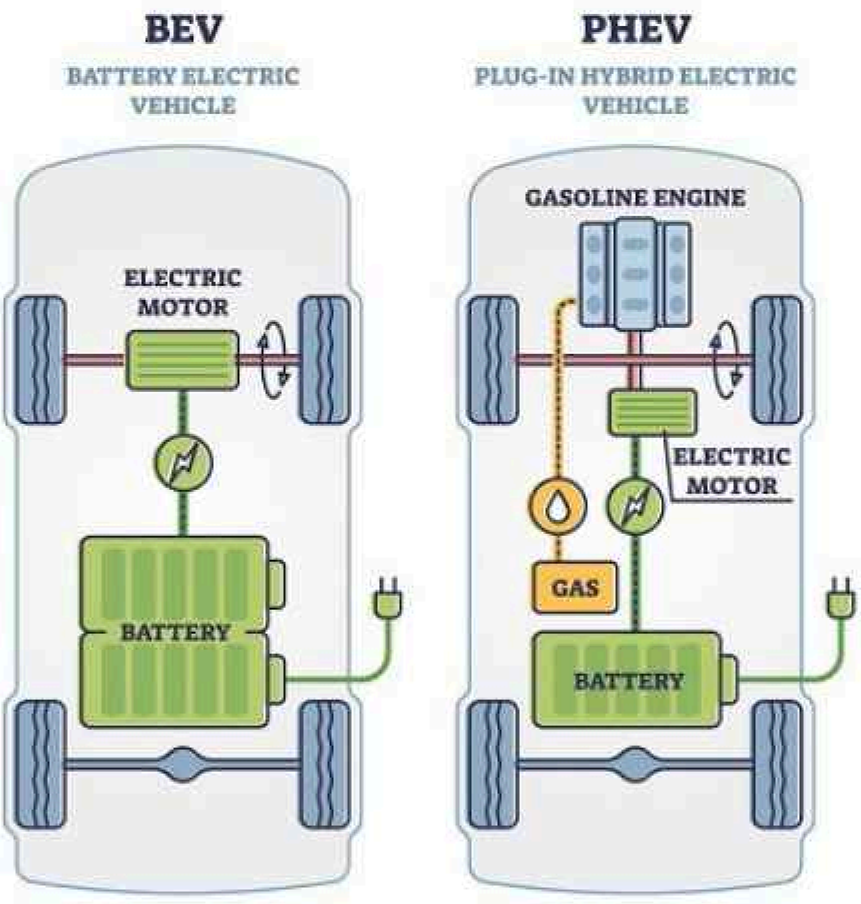
key features and benefits of solid-state batteries:

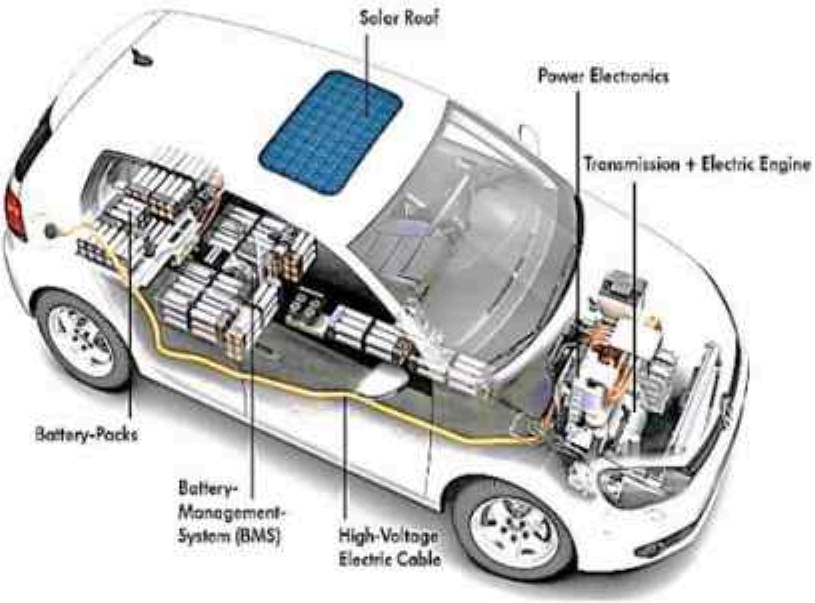
1. Improved Energy Density:
   - Solid-state batteries use solid electrolytes instead of liquid electrolytes found in traditional lithium-ion batteries. This design allows for higher energy density, potentially providing electric vehicles with increased range on a single charge.
2. Enhanced Safety:
   - Solid-state batteries are considered safer than traditional lithium-ion batteries because they eliminate the use of flammable liquid electrolytes. This reduces the risk of fire and enhances overall safety in electric vehicles.
3. Faster Charging Rates:
   - Solid-state batteries have the potential to offer faster charging rates compared to conventional lithium-ion batteries. The absence of liquid electrolytes and the improved conductivity of solid electrolytes contribute to quicker charging times.
4. Extended Lifespan:
   - The solid-state battery design can contribute to a longer lifespan for electric vehicle batteries. This is advantageous for both consumers and the environment, as it reduces the frequency of battery replacements and associated waste.
5. Temperature Tolerance:
   - Solid-state batteries often exhibit better temperature tolerance, allowing electric vehicles to perform well in a wider range of environmental conditions. This is particularly beneficial in extreme temperatures where battery performance can be a concern.
6. Reduced Size and Weight:
   - Solid-state batteries may enable the design of more compact and lightweight battery packs for electric vehicles. This can contribute to improved overall vehicle efficiency and design flexibility.
7. Compatibility with Advanced Materials:
   - Solid-state battery technology is compatible with advanced electrode materials, which can further enhance performance and energy storage capabilities. This opens up possibilities for continued improvements in battery technology.

Electric Vehicle Types

Electric vehicles (EVs) come in various types, each designed to meet specific needs and preferences. The main types of electric vehicles include:

1. Battery Electric Vehicles (BEVs):
   - BEVs are fully electric vehicles that run solely on electric power stored in batteries. They do not have an internal combustion engine (ICE) and produce zero tailpipe emissions. BEVs are charged by plugging into an electric power source, typically at home, charging stations, or charging infrastructure.
2. Plug-in Hybrid Electric Vehicles (PHEVs):
   - PHEVs have both an electric motor and an internal combustion engine. They can operate in an all-electric mode for a certain range before switching to the internal combustion engine when the battery is depleted. PHEVs can be charged through a power outlet, offering flexibility for electric-only driving and extended range with the gasoline engine.
3. Hybrid Electric Vehicles (HEVs):
   - HEVs combine an internal combustion engine with an electric motor. Unlike PHEVs, HEVs cannot be charged externally; instead, they rely on regenerative braking and the internal combustion engine to charge the battery. The electric motor assists the engine to improve fuel efficiency and reduce emissions.
4. Extended-Range Electric Vehicles (EREVs):
   - EREVs are similar to PHEVs but are designed to primarily operate on electric power. The internal combustion engine serves as a generator to recharge the battery when needed, extending the overall range of the vehicle. The electric motor provides propulsion, and the engine is only used when the battery is depleted.
5. Fuel Cell Electric Vehicles (FCEVs):
   - FCEVs use a fuel cell to generate electricity through a chemical reaction between hydrogen and oxygen. This electricity powers an electric motor, and the only byproduct is water vapor. FCEVs are an alternative to battery electric vehicles, offering longer driving ranges and relatively quick refueling times. However, they currently face challenges related to infrastructure and hydrogen production.
6. Micro Electric Vehicles (Micro EVs):
   - Micro EVs are compact, lightweight electric vehicles designed for short-distance urban commuting. They are often smaller than traditional cars and may include electric scooters, electric bicycles, and small electric cars. These vehicles are ideal for navigating congested city streets.
7. Electric Buses and Trucks:
   - Electric buses and trucks are designed for public transportation and freight delivery. They use electric powertrains to reduce emissions and operational costs. Electric buses are increasingly adopted in urban transit systems, while electric trucks aim to address environmental concerns in the freight industry.
8. Autonomous Electric Vehicles (AEVs):
   - AEVs combine electric propulsion with autonomous driving technology. These vehicles use sensors and advanced control systems to navigate without human intervention. AEVs are at the forefront of innovation in the automotive industry, aiming to revolutionize transportation and improve safety.





LAKSHMI M, IV year EEE

Recent Advancements in IoT

1. Edge Computing in IoT:
   - Edge computing has gained prominence in IoT to process data closer to the source, reducing latency and improving real-time decision-making. This is particularly crucial for applications requiring low-latency responses, such as industrial IoT and autonomous vehicles.
2. 5G Integration:
   - The rollout of 5G networks has significantly impacted IoT by providing faster and more reliable connectivity. The high data transfer rates and low latency of 5G enable IoT devices to communicate more efficiently, supporting applications like smart cities, healthcare, and industrial automation.
3. AI and Machine Learning in IoT:
   - Integration of artificial intelligence (AI) and machine learning (ML) algorithms within IoT systems enhances data analysis, predictive modeling, and automation. This enables IoT devices to learn and adapt to changing conditions, improving overall system efficiency.
4. Blockchain for IoT Security:
   - Blockchain technology is being applied to enhance the security of IoT devices and data. It provides a decentralized and tamper-resistant way of recording transactions, ensuring the integrity and confidentiality of data in IoT ecosystems.
5. Digital Twins:
   - Digital twins, virtual representations of physical objects or systems, are increasingly used in IoT for monitoring, analysis, and simulation. This technology allows organizations to create a digital replica of physical assets, providing insights into performance, maintenance needs, and optimization opportunities.
6. IoT Interoperability Standards:
   - Efforts to establish common IoT standards and protocols are ongoing to improve interoperability among devices and platforms. Standardization helps facilitate seamless communication and integration across diverse IoT ecosystems.
7. IoT Security Enhancements:
   - With the growing number of connected devices, there is an increased focus on enhancing IoT security. This includes the development of secure device provisioning, authentication methods, and encryption protocols to protect against cyber threats.
8. Energy Harvesting for IoT Devices:
   - Innovations in energy harvesting technologies are enabling the development of IoT devices that can generate their own power from ambient sources, such as solar, vibration, or thermal energy. This reduces the reliance on traditional power sources and extends device lifespan.
9. IoT in Healthcare:
   - The healthcare sector has seen advancements in IoT applications, including remote patient monitoring, wearable health devices, and smart medical equipment. These technologies aim to improve patient care, enhance efficiency, and enable better health outcomes.
10. Smart Agriculture:
    - IoT is playing a significant role in agriculture with the deployment of sensors, drones, and monitoring systems. These technologies help farmers optimize crop yields, conserve resources, and make data-driven decisions.
11. IoT in Supply Chain Management:
    - IoT is being increasingly adopted in supply chain management for tracking and monitoring the movement of goods. This includes the use of RFID tags, sensors, and GPS technology to provide real-time visibility into the supply chain, improving efficiency and reducing operational costs.

These trends represent a snapshot of the ongoing advancements in the IoT landscape. For the latest developments, it is recommended to refer to recent publications, industry reports, and news sources that provide up-to-date information on IoT innovations.

DIVYAJEYASHREE V, IV year EEE

02.

Arduino Programming

Writing Your First Sketch (Program):

- Open the Arduino IDE.
- Write your first program, known as a "sketch," in the Arduino IDE. The basic structure of an Arduino sketch includes setup() and loop() functions.

```
void setup() {
  // Code to run once when the sketch starts
}

void loop() {
  // Code to run repeatedly in a loop
}
```

```
int ledPin = 13;

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  digitalWrite(ledPin, HIGH);
  delay(1000);
  digitalWrite(ledPin, LOW);
  delay(1000);
}
```

Digital and Analog I/O:

- Use pinMode() to set a pin as input or output.
- Use digitalWrite() to write a digital HIGH or LOW value to a pin.
- Use digitalRead() to read the digital value from a pin.
- For analog sensors, use analogRead() to read analog values (0-1023) from analog pins.

Serial Communication:

- Use Serial.begin() to initialize serial communication.
- Use Serial.print() and Serial.println() to send data to the Serial Monitor for debugging.

```
void setup() {
  Serial.begin(9600);
}

void loop() {
  Serial.println("Hello, Arduino!");
  delay(1000);
}
```

```
void blinkLED(int pin, int duration)
{
  digitalWrite(pin, HIGH);
  delay(duration);
  digitalWrite(pin, LOW);
  delay(duration);
}
void setup() {
  pinMode(13, OUTPUT);
}
void loop() {
  blinkLED(13, 500);
}
```

Functions and Libraries:

Define custom functions for modular code.
Explore and use Arduino libraries to simplify complex tasks.

KALIRAJ R, III year EEE

AUGMENTED REALITY (AR) AND VIRTUAL REALITY (VR)

1. Definition:
- AR overlays digital content onto the real-world environment, enhancing the user's perception of the surroundings.

2. Interaction:
- AR allows users to interact with both the physical and virtual elements simultaneously. Digital information, such as graphics, text, or 3D models, is superimposed onto the real world.

3. Devices:
- AR experiences can be delivered through various devices, including smartphones, tablets, smart glasses, and heads-up displays. Popular examples include AR apps on smartphones and AR glasses like Microsoft HoloLens.

4. Applications:
- AR finds applications in diverse fields such as gaming, navigation, education, healthcare, and manufacturing. It enhances real-world scenarios by providing contextually relevant information.

1. Definition:
   - VR creates a fully immersive digital environment that isolates users from the real world, replacing it with a computer-generated, interactive environment.
2. Interaction:
   - In VR, users are entirely immersed in the digital environment, often using headsets or goggles. Interaction is primarily within the virtual space, and users may use controllers or other input devices.
3. Devices:
   - VR experiences are typically delivered through VR headsets or goggles, such as Oculus Rift, HTC Vive, or PlayStation VR. These devices provide a closed-off, immersive experience that isolates users from the external environment.
4. Applications:
   - VR is used in gaming, simulations, training, education, healthcare, and virtual tourism. It allows users to explore and interact with entirely synthetic environments.

JEYAPADMA B, II year EEE

Dual Axis Solar Power Tracker System

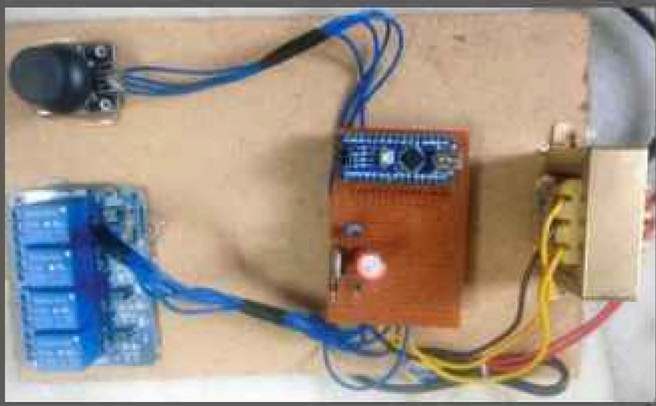
RAJKUMAR M, II Year EEE B section

This system requires the involvement of a wide range of engineering including mechanical, electrical, and electronics. The mechanical part would involve designing a smooth gear system to move as per requirement. The electrical part would be the working of solar panel and battery requirement.
The electronics would involve designing the sensor system that would generate commands for the gear system to act accordingly. The system employs spur gear for the implementation of the dual-axis solar tracker. The system is implemented using Atmel IC AT89C51.
This electrical project is available at Dual Axis solar tracker system.

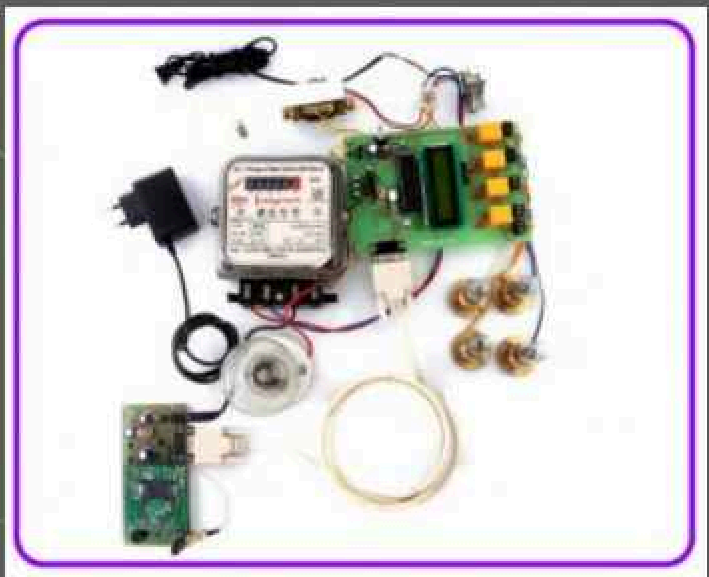
Joystick-Controlled Industrial Automation System

When joystick shaft is in normal position, it is in standby mode. When it is moved along horizontal axis (right and left), voltage at VRx pin changes. Similarly, when it is moved along vertical axis (up and down), voltage at VRy pin changes. So, we have four different output signals for four directions of the joystick, which are connected to two analogue-to-digital converter (ADC) input pins A0 and A1 of Arduino Nano.
When the shaft is moved, voltage on each pin goes high or low depending on direction. The joystick module also comes with one push-button switch for interruption purpose (this function is not used here).

ARUNKUMAR.S, II year EEE - A section



PRAVEEN SELVAMANI M, II year EEE - B sec.



Energy Meter Billing with Load Control over GSM with User Programmable Number Features

Now a day the electricity department hiring employees every month to take meter readings in every home, which is a very expensive and time-consuming job. Thus, this system provides a convenient and efficient method to avoid this problem. This project is used to receive the monthly energy consumption from a remote area straight to an electrical department as well as to the consumers and also to control the loads through SMS using GSM technology. This is an electrical engineering project but can also do electronic engineering students.

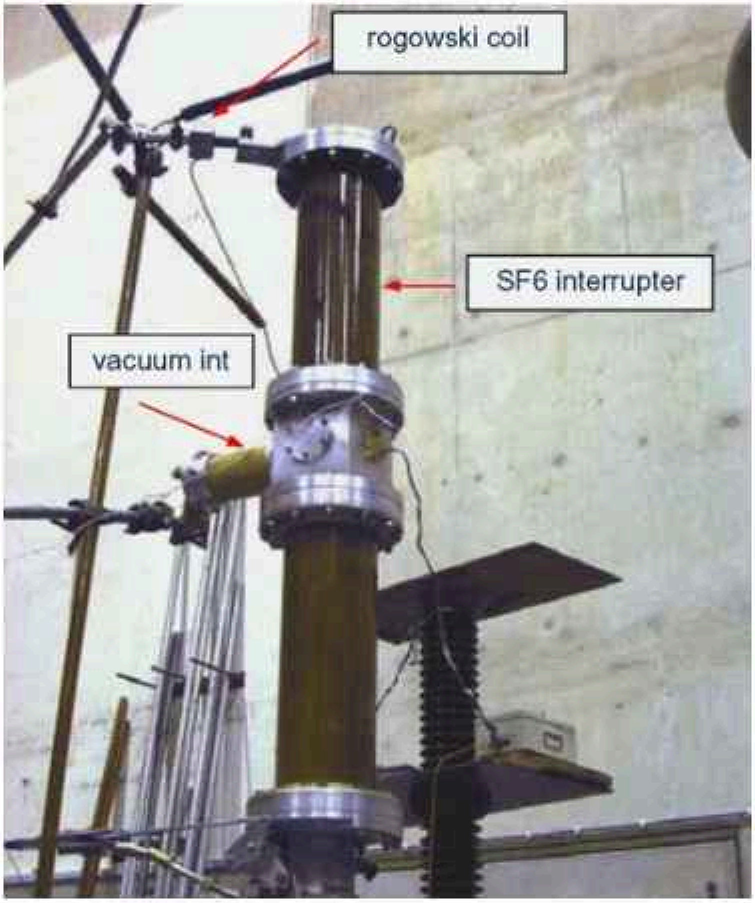
Hybrid (Vacuum-Gas) High Voltage Circuit Breaker

Hybrid gas circuit breakers combine the advantages of traditional gas-insulated circuit breakers with eco-friendly insulating gases. These breakers aim to reduce the environmental impact while maintaining high performance and reliability.

A hybrid circuit breaker consists of a series connected vacuum- and SF6(CO2) interrupter. The vacuum interrupter has the function to withstand very steep-rising transient recovery voltages, whereas the SF6(now CO2) interrupter is stressed with the peak of it.

The interaction between the two arcs has a positive effect on the interruption: Immediately before current zero, the SF6(CO2) arc assists the vacuum arc to interrupt, whereas after current zero, the vacuum arc assists the SF6 arc in the recovery against the recovery voltage.

A prototype hybrid CB for 145 kV/ 63 KA show under the test in pic.



Summary of hybrid circuit breakers benefits:

- Development in vacuum technology, particularly very high short-circuit current interrupting capability that has been obtained in bottles of reduced size.
- No environmental problem of SF6 gas by using alternative gas as CO2.
- Function at low ambient temp and a high interrupting capacity without additional capacitor

ALAGARSAMY YASVANTH T R, II year EEE, A section

EVERYONE WANTS TO LIVE ON TOP OF THE MOUNTAIN BUT ALL THE HAPPINESS AND GROWTH OCCURS WHILE YOU'RE CLIMBING IT.

ANDY ROONEY

MAY 2022

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